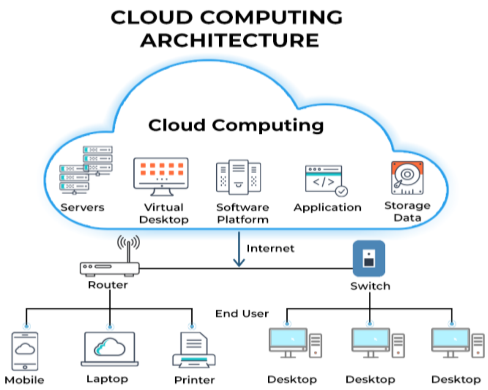
Unit - I
Fundamentals of Cloud Computing

➢ Cloud computing

Cloud computing is the on-demand delivery of IT resources, like storage, databases, and processing power, over the internet, allowing users to access them from anywhere and pay only for what they use.
Instead of buying and maintaining physical hardware, users access these services from cloud providers on a rental basis.
It similar to how you use services like Google Drive, Netflix, or Gmail, which store your data and run applications on the provider's servers rather than your own computer.
The cloud computing model gives users, greater flexibility and scalability.



➢ Characteristics of Cloud Computing

1. On-Demand Self-Services:

Users can independantly access computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with the service provider.

2. Broad Network Access:

The Computing services are available over the network and accessed through standard mechanisms, so that it can be used by various client platforms like mobile phones, tablets, and laptops.

3. Resource Pooling:

A cloud service provider can provide each client with different services based on their demands by employing resource pooling to divide resources across many clients.
Resource pooling is a multi-client approach for location independence where the customer generally has no knowledge over the exact location of provided resource.

4. Rapid Elasticity:

Capabilities can be rapidly and elastically provisioned on a need basis.
Whenever the user require services it is provided to him and it is scale out as soon as its requirement gets over.

5. Measured Service:

The resource utilization is tracked for each application and tenant, it will provide both the user and the resource provider with an account of what has been utilized.
This is done for various reasons like monitoring billing and effective use of resource.

Benefits of Cloud Computing

Cost Savings

Without having to put capital investment for significant IT infrastructure and personnel to manage it, cloud computing enables you to pay for what you use.

Scalability and Elasticity

Cloud provides massive amount of computing resources.
It allows businesses to quickly scale computing resources up or down to meet demand.

Flexibility and Agility

Enables businesses to adapt quickly to changing needs and bring products to market faster.
Provides on-demand access to a vast range of services and technologies.

Improved Accessibility

Data and applications can be accessed from anywhere with an internet connection, promoting mobility and remote work.

Enhanced Collaboration

Facilitates teamwork among widely distributed teams through shared access to data and applications.

Automatic Updates and Maintenance

Cloud providers handle software updates, giving users access to the latest features and technologies without manual intervention.

Disaster Recovery and Data Loss Prevention

Provides built-in redundancy and backup solutions to protect against data loss and ensure business continuity.

Security

Cloud providers often offer advanced security measures and centralized data security.

Cloud Deployment Models

A cloud deployment model defines ***where cloud resources are hosted***, **who manages them**, and **how they are accessed**.

The main deployment models are :-

- Public Cloud
- Private Cloud
- Hybrid Cloud
- Community Cloud

1. Public Cloud

A public cloud is a cloud computing service where a third-party provider owns, operates, and manages **computing resources** like servers, storage, and networks over the internet, making them available to use by general public.

It is not tied to particular organization.

Customers access these services through the internet, pay for what they use.

It is provider's responsibility for maintenance and security.

Well-known examples include Amazon Web Sevice, Microsoft Azure and Google Cloud.

Advantages of the Public Cloud Model

- **Cost Efficient**:
  In the public cloud, we have to pay for what we used.
  So it is more cost-efficient than maintaining the physical servers or their own infrastructure.
- **Infrastructure Management is not required:**
  Using the public cloud does not necessitate infrastructure management.
- **No maintenance:**
  The maintenance work is done by the service provider (not users).
- **Dynamic Scalability:**
  To fulfill your company's needs, on-demand resources are accessible.
- **Automatic Software Updates:**
  In the public cloud, there are automatic software updates. We don't have to update the software manually.
- **Accessibility:**
  Public clouds allow users to access their resources and applications from anywhere in the world. We just need an internet connection to access it.

Disadvantages of the Public Cloud Model

- **Less secure:** Public cloud is less secure as resources are public so there is no guarantee of high-level security.
- **Low customization:** It is accessed by many public so it can't be customized according to personal requirements.
- **Limited Control:** With public cloud services, users have limited control over the infrastructure and resources used to run their applications.
- **Service Downtime:** Public cloud providers may experience service downtime due to hardware failures, software issues, or maintenance activities. This can result in temporary loss of access to applications and data.
- **Connectivity requirements :** Public cloud services require a reliable and stable internet connection to access the resources and applications hosted in the cloud. If the internet connection is slow or unstable, it can affect the performance and availability of the services.

2. Private Cloud

Private Cloud is a cloud environment used exclusively by a **single organization** for example, a company or government, and it is **accessed** by **only authorized users** within that **organization**. It can be hosted on-premises within the organization's data center or managed by a third-party provider.

Advantages:-

Enhanced Security & Privacy:
Private clouds provide a higher level of data security and privacy because resources are dedicated to a single organization, reducing the risk of unauthorized access.

Greater Control:
Private cloud users have full control over the infrastructure and services, which means they can customize the environment to their specific needs.

Improved Performance:
Organizations with dedicated resources and control over the network infrastructure can often achieve better performance and lower latency compared to public clouds. This is particularly important for applications that require fast response times and high throughput.

Customization to business needs:
Private clouds offer a high degree of customisation. This level of customisation is often not possible in public cloud environments.

Private Cloud Disadvantages:

Higher Cost: Private cloud services can be more expensive than public cloud services since they require a dedicated infrastructure including hardware, software, and networking.

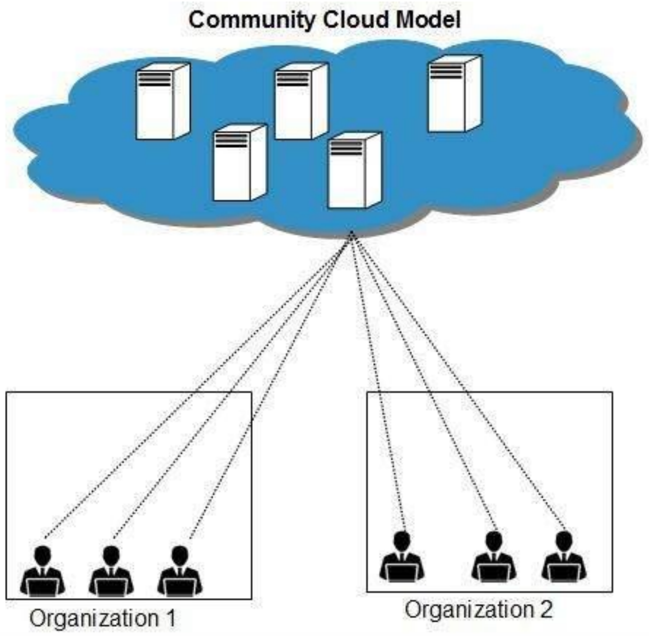
Maintenance: Private cloud users are responsible for maintaining and updating the infrastructure, which can be time-consuming and costly.

Limited Scalability: Private cloud services can be less scalable than public cloud services, which means they may not be able to handle sudden spikes in demand. For example, if it has decided to use 5TB storage initially, it can not dynamicaly consume 6 TB.

Inadequate Resource Utilization:-
Organizations may not fully utilize their dedicated resources, leading to wasted infrastructure and under utilization of resourcse.

3. Community cloud

Community cloud is a cloud computing model where infrastructure and resources are shared by several organizations within a specific community that share common interests, such as industry-specific regulations, mission, or project goals
It allows system and services to be **accessible** by **group of organizations.**
It may be managed **internally** by **organizations** or by the **third-party**.
Example Many banks such as ICICI, HDFC, AXIX etc. can come together and build a community cloud, that can serve their respective requirements.
The Community Cloud Model is shown in the diagram below.



Advantages : -

- **Shared Investment :**
  By sharing the cost of infrastructure and resources, community clouds **significantly reduce the financial burden** on individual organizations compared to building and maintaining their own private clouds.

- **Controlled Access and Security:**
  The shared nature of community clouds allows for **controlled access** and **data sharing**, **enhancing security** and **privacy** compared to public clouds.

- **Easy to meet Industry-Specific Compliance:**
  Community clouds are designed specifically for particular group of Organizations , so it is **easier** to **achieve** specific security and compliance requirements of a particular industry, such as healthcare or finance.

- **Adaptable to Changing Needs (Scalability):**
  Community clouds offer scalability, allowing organizations to adjust their resource usage based on demand and easily scale up or down as needed.

Disadvantages : -

- **Lower trust and Adoption :-**
  Since the organizations that form the community are also business competitors, there is in general lower trust issue among them. Hence the adoption of community cloud might be lower.

- **Potential Conflicts:**
  Different organizations within the community cloud may have varying needs and priorities, which may lead to conflicts and delays in decision-making.
  For example, one organization might prioritize cost savings while another prioritizes data security.

- **Limited Customization:**
  Community clouds are shared environments, hence individual organizations have limited degree of customization and control over infrastructure as compared to private clouds.

- **Higher Costs (Potentially):**
  While community clouds can be more cost-effective than private clouds, the shared cost model can be problematic if some organizations require more resources or have higher security needs. Cost-sharing arrangements need to be clearly defined and fair to avoid dissatisfaction.

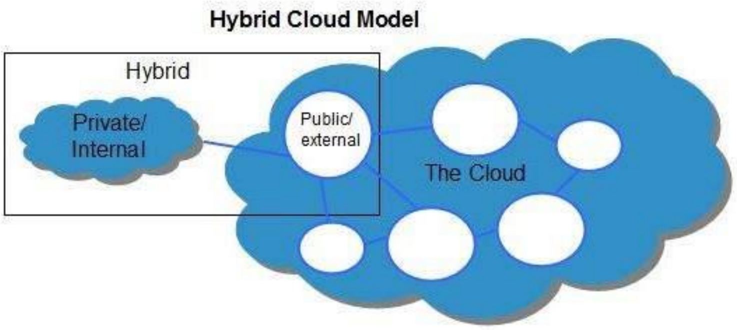
- **Limited Scalability:**
  Community clouds may have limited scalability compared to public clouds because they share resources among a specific group of organizations.

- **Security Concerns:**
  Security can be a major concern, as different organizations may have different security requirements and standards, potentially leading to vulnerabilities.

4. Hybrid Cloud

A hybrid cloud is a combination of public cloud and a private cloud.
This approach allows organizations to achieve the benefits of both public and private clouds.
Using Hybrid model, organizations can gain the flexibility and scalability of public clouds along with the security, control of private infrastructure.
Public cloud resources provide the ability to scale up quickly, while private clouds offer more control over sensitive data and critical workloads.
Data and applications can move between the private and public clouds, enabling companies to optimize resources.

Example :- Airbnb uses a public cloud for its website, mobile app, and analytics, while keeping sensitive user data like payment information on its private cloud infrastructure.



Advantages :-

1. Flexibility and Scalability:
Hybrid clouds provide the flexibility to choose the optimal environment for different workloads. Non-sensitive tasks can be moved to the public cloud for cost-effectiveness and scalability, while critical data and applications can remain on-premises for enhanced security and control.

2. Enhanced Security :
It provides security by keeping sensitive data in a private cloud.

3. Cost Optimization:
Cost of hybrid clouds can be significantly reduced, as it takes the advantage of scalability and pay-as-you-go model of public clouds for certain workloads

4. Improved Disaster Recovery and Business Continuity:
The hybrid cloud model provide better disaster recovery by allowing organizations to replicate data and workloads across public and private clouds.
In the event of an outage, they can easily switch to the cloud environment, minimizing downtime and ensuring business operations continue without interruption.

6. Increased Agility and Innovation:
By combining the flexibility and scalability of public clouds with the security and control of private clouds, organizations can accelerate innovation and respond quickly to changing business needs.

Disadvantages :-

Complexity:
Managing a hybrid cloud environment requires specialized expertise to integrate, monitor, and maintain seamless communication between private and public clouds. This complexity can be challenging.

Higher Costs:

While public cloud services offer pay-as-you-go models, setting up a private cloud infrastructure within a hybrid model involves significant capital investment in hardware, software, and personnel.

Dependency on Internet Connection:
As with public cloud, hybrid cloud deployments rely on a stable internet connection.
A disruption in the connection can interrupt access to remotely stored data and impact business operations

Cloud Services Models

The main cloud services models are
Infrastructure as a Service (IaaS)
Platform as a Service (PaaS)
Software as a Service (SaaS)
Each model offering different levels of management and control for users.

Software as a Service (SaaS)

Software as a Service (SaaS) is a cloud computing model where a provider hosts software and makes it available to customers over the internet, typically on a subscription basis.
(SaaS) means using software over the internet instead of installing in on your computer. You don't have to worry about downloading, updating, or maintaining anything- the company that provides the software handles all of that.
Users access the software through a thin client interface such as web browser.
Popular examples include Google Docs, Zoom, and Microsoft 365.
Think of Google Docs. You don't need to install it. You just open your browser, log in, and start using it.
Google stores your work and keeps the software updated. You just use it when you need it.

Advantages of SaaS

- **Cost-Effectiveness:**Reduces upfront costs because you pay a subscription rather than buying the software.
- **Automatic Updates and Maintenance:**The provider handles all software updates and maintenance, saving your organization time and resources.
- **Scalability:**Easily scales resources up or down based on business needs, allowing for adjustments as the company grows or shrinks.
- **Accessibility:**Accessible from any location and any connected device, facilitating remote work and collaboration.
- **No Hardware/Installation Hassles:**Eliminates the need to purchase, install, or maintain hardware and software, as the provider manages the infrastructure.
- **Faster Deployment:**Quick to deploy and commission since there are no lengthy installation processes.

Disadvantages of SaaS

- **Loss of Control:**You have less control over the software, as the provider manages updates and changes that you must adopt.
- **Security and Data Concerns:**Data is stored on third-party servers, which raises security concerns and can make it difficult to follow data protection regulations.
- **Internet Dependency:**Performance is heavily dependent on a reliable internet connection, and lack of connectivity can hinder usability.
- **Customization Limits:**Customization options are often limited to what the provider offers, leading to less flexibility for specific needs.
- **Integration Challenges:**Integrating SaaS applications with existing in-house software can sometimes be difficult and lead to compatibility issues.
- **Vendor Lock-In:**It can be challenging to switch providers or migrate data to a different system once you are deeply integrated with a SaaS vendor.

Platform as a Service (PaaS)

Cloud PaaS, or Platform as a Service, is a cloud computing model that provides a complete development and deployment environment over the internet,
It allows developers to build, run, and manage applications without handling the underlying infrastructure like servers, operating systems, and networking..
PaaS includes pre-built software components, development tools, operating systems, and middleware, enabling faster software development and simplified deployment while a cloud provider manages the infrastructure.

Examples :- Google App Engine, and Azure App Service.